

KNN Classification With One-Step Computation

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Abstract—KNN classification is an improvisational learning mode, in which they are carried out only when a test data is predicted that set a suitable K value and search the K nearest neighbors from the whole training sample space, referred them to the lazy part of KNN classification. This lazy part has been the bottleneck problem of applying KNN classification due to the complete search of K nearest neighbors. In this paper, a one-step computation is proposed to replace the lazy part of KNN classification. The one-step computation actually transforms the lazy part to a matrix computation as follows. Given a test data, training samples are first applied to fit the test data with the least squares loss function. And then, a relationship matrix is generated by weighting all training samples according to their influence on the test data. Finally, a group lasso is employed to perform sparse learning of the relationship matrix. In this way, setting K value and searching K nearest neighbors are both integrated to a unified computation. In addition, a new classification rule is proposed for improving the performance of one-step KNN classification. The proposed approach is experimentally evaluated, and demonstrated that the one-step KNN classification is efficient and promising.

Index Terms—KNN, setting K value, searching K nearest neighbors, group lasso

1 INTRODUCTION

KNN classification is one of top 10 algorithms in data mining [1]. It has widely and successfully been applied to many areas, such as text classification [2], pattern recognition [3] and image processing [4]. Standard KNN classification predicts a test data with only its K nearest neighbors in the training dataset. One need only select/construct a distance function and a classification rule when designing his/her KNN classification algorithm [5]. A KNN classification algorithm is a query triggered yet improvisational learning procedure, in which they are carried out only when a test data is predicted that set a suitable K value and search the K nearest neighbors from the whole training sample space. This means KNN classification includes a lazy part, setting K value and searching K nearest neighbors. From the learning procedure, designing a KNN classification algorithm is almost independent of the training data, i.e., it does not train anything with the training dataset before a query is submitted. Therefore, KNN classification is attributed to a non-model, or nonparametric, or lazy classification method [6].

To deal with the lazy part of KNN classification, there are recently many efforts to setting K value or searching K nearest neighbors. For example, Gora and Wojna presented an algorithm of learning the optimal-K-values for a test data [7]. Guo *et al.* advocated to automatically seek the optimal-K-value for a test data [8]. Wang *et al.* argued to locally adjust the number of nearest neighbors [9]. Manocha *et al.* designed a probabilistic method for computing the optimal-K-values

of nearest neighbors [10]. Li *et al.* spelled out why different categories need different numbers of nearest neighbors [11]. Sun *et al.* designed an adaptive algorithm of finding the optimal-K-values for a test data [12]. Zhang *et al.* advocated to reconstruct the relation matrix and built a optimal-K-value computation [5]. Gou *et al.* proposed a K-nearest neighbor algorithm by local mean vector representation [13]. Pan *et al.* proposed an adaptive KNN algorithm, which not only considers the majority class, but also considers the second majority class in the K nearest neighbors of a test data [14].

However, the above KNN classification algorithms are still lazy. Recently, a non-lazy KNN classification has been developed in [15], denoted as Ktree and K*tree classification algorithms. Ktree is designed for fast finding the K value of a test data. It adds a training stage to traditional KNN method and thus outputting a training model, i.e., building a decision tree, Ktree, to predict the optimal-K-values for each test data. K*tree is proposed for both setting K value or searching K nearest neighbors. It is an extension of Ktree by appending sets of nearest neighbors of training samples in each leaf node. In this way, the K*tree KNN classification is not lazy.

From these reports, setting K value and searching K nearest neighbors are carried out in two phases. In general, searching K nearest neighbors is a complete training sample space search, i.e., it is an NP problem [16]. Although the K*tree KNN classification is not lazy, the lazy part of KNN classification is also accomplished in two steps as follows. For a test data, the first step searches its nearest neighbor X and the K value. The second step identifies the K-1 nearest neighbors from the set of nearest neighbors of X.

To attack the above issues in KNN classification, a one-step KNN classification algorithm is designed in this article that reduces the complexity of searching K nearest neighbors to $O(N^2)$. The one-step KNN classification is a one-step computation that integrates both setting K value and searching K nearest neighbors to a unified computation. Specifically, given a test data, training samples are first applied to fit the test data with the least squares loss function. And then, a relationship matrix is generated by

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